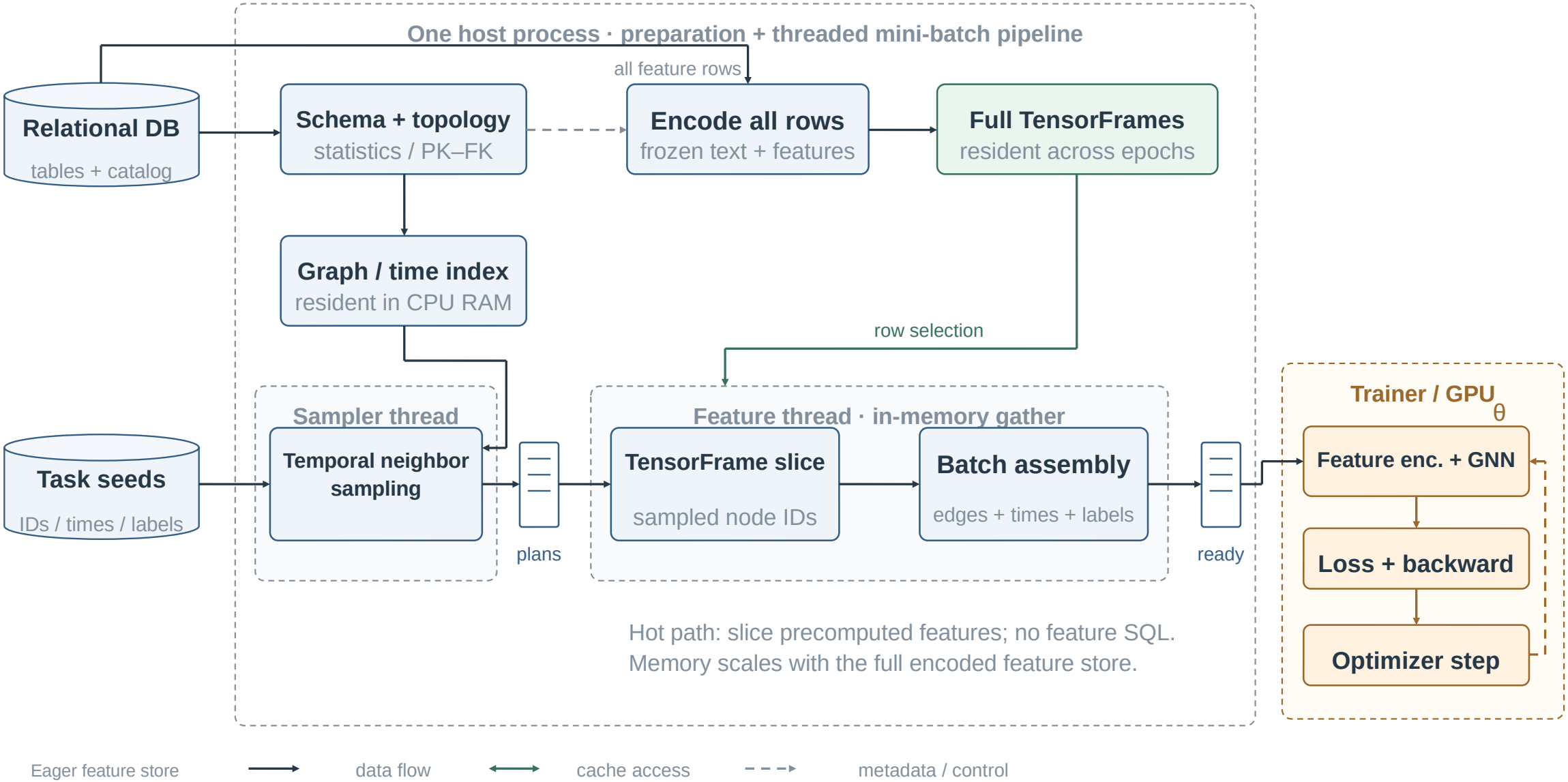
(a) Batch ｜全量物化训练架构

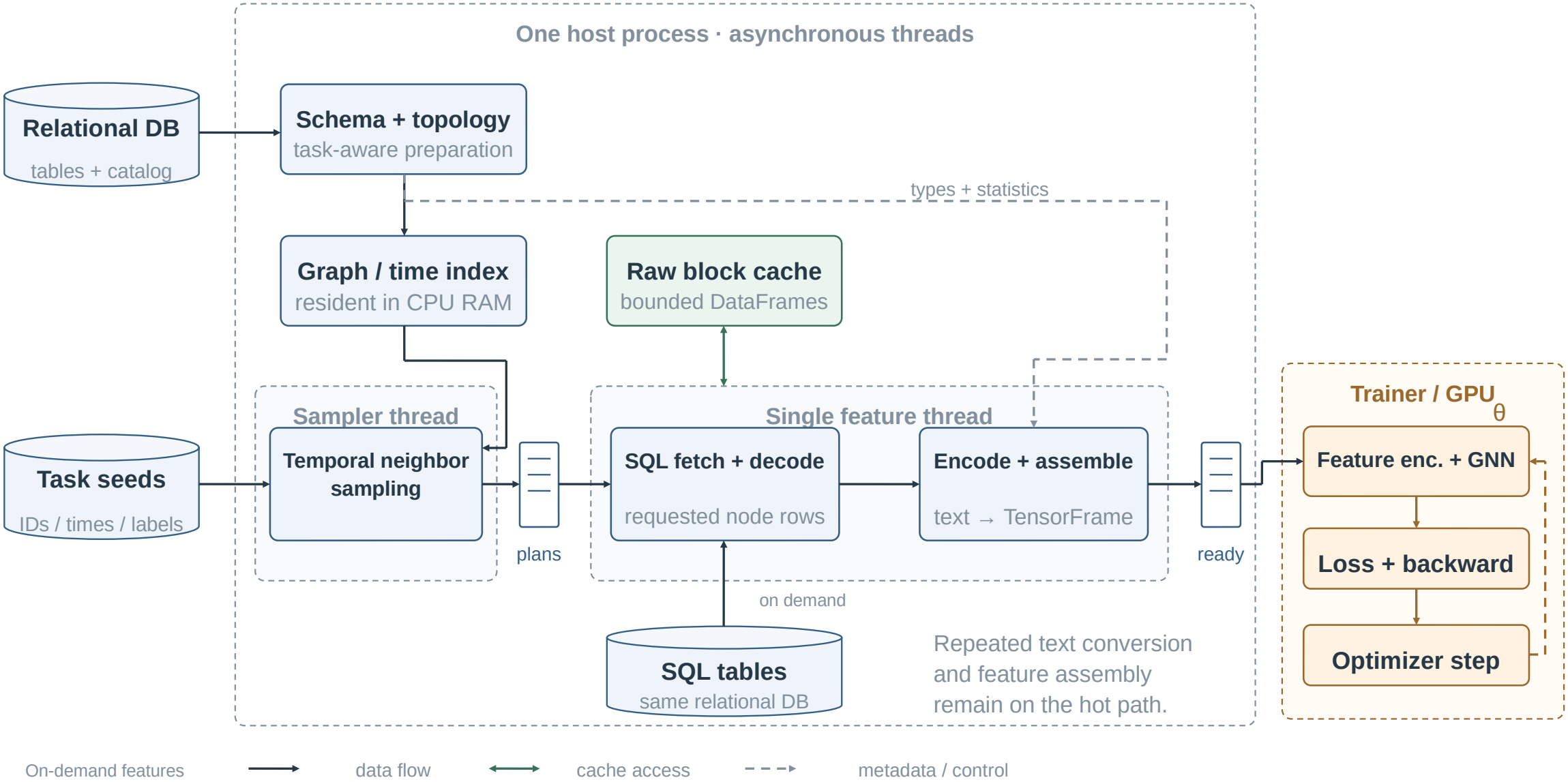
训练前生成全部特征；训练时从常驻 TensorFrame 切片。虚线框表示执行范围。



Batch: eager materialization followed by sampled mini-batch training. Static features are resident; model parameters remain trainable.

(b) Vanilla ｜按需读取的在线训练架构

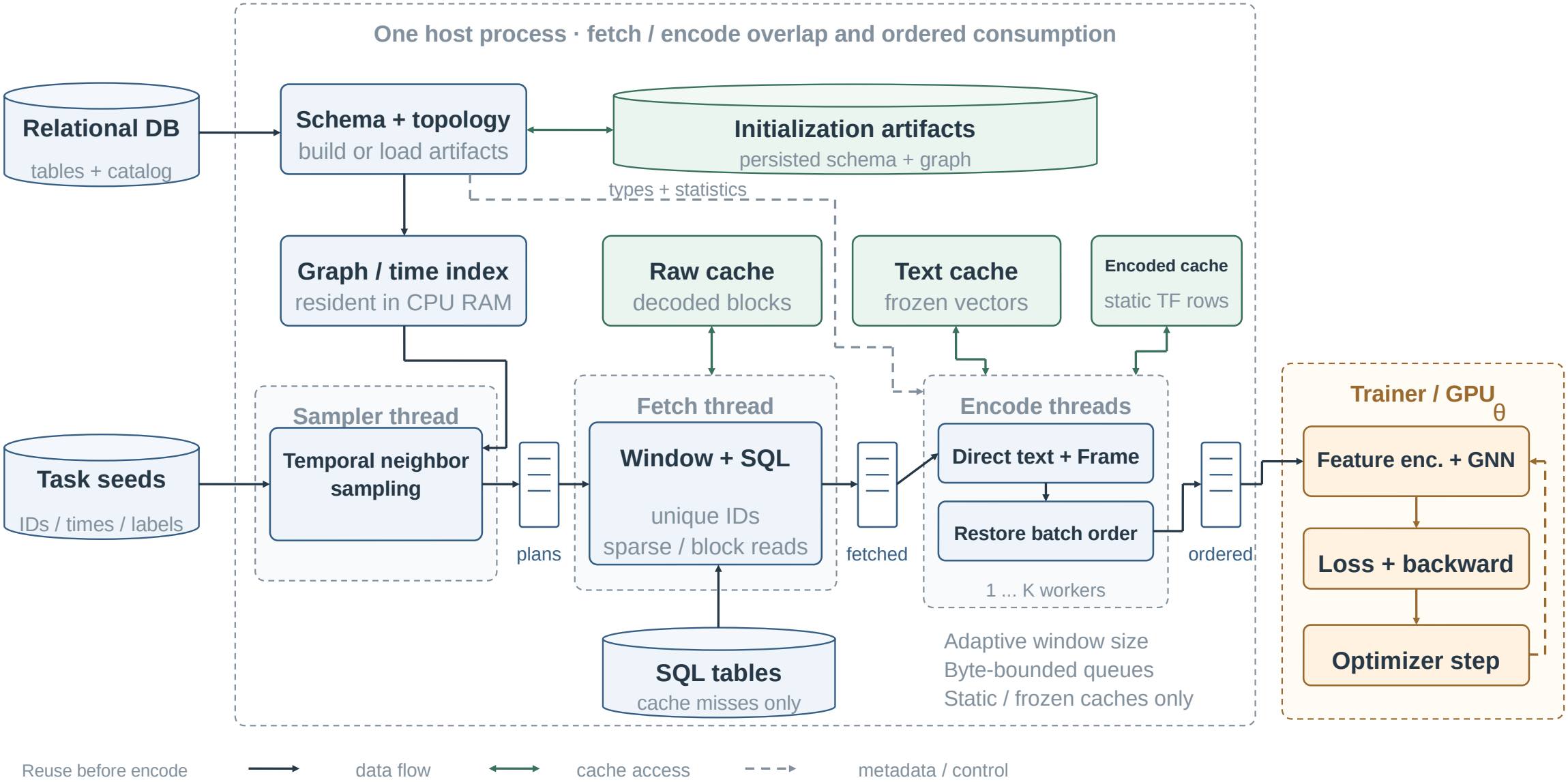
单进程内异步采样、特征线程与训练；已有 raw cache ，但缺少后续文本与编码特征复用。



Vanilla: bounded online feature access with a raw cache and asynchronous threads; static text / encoded-feature reuse is not yet added.

(c) v1 ｜单进程优化：窗口复用与分层缓存

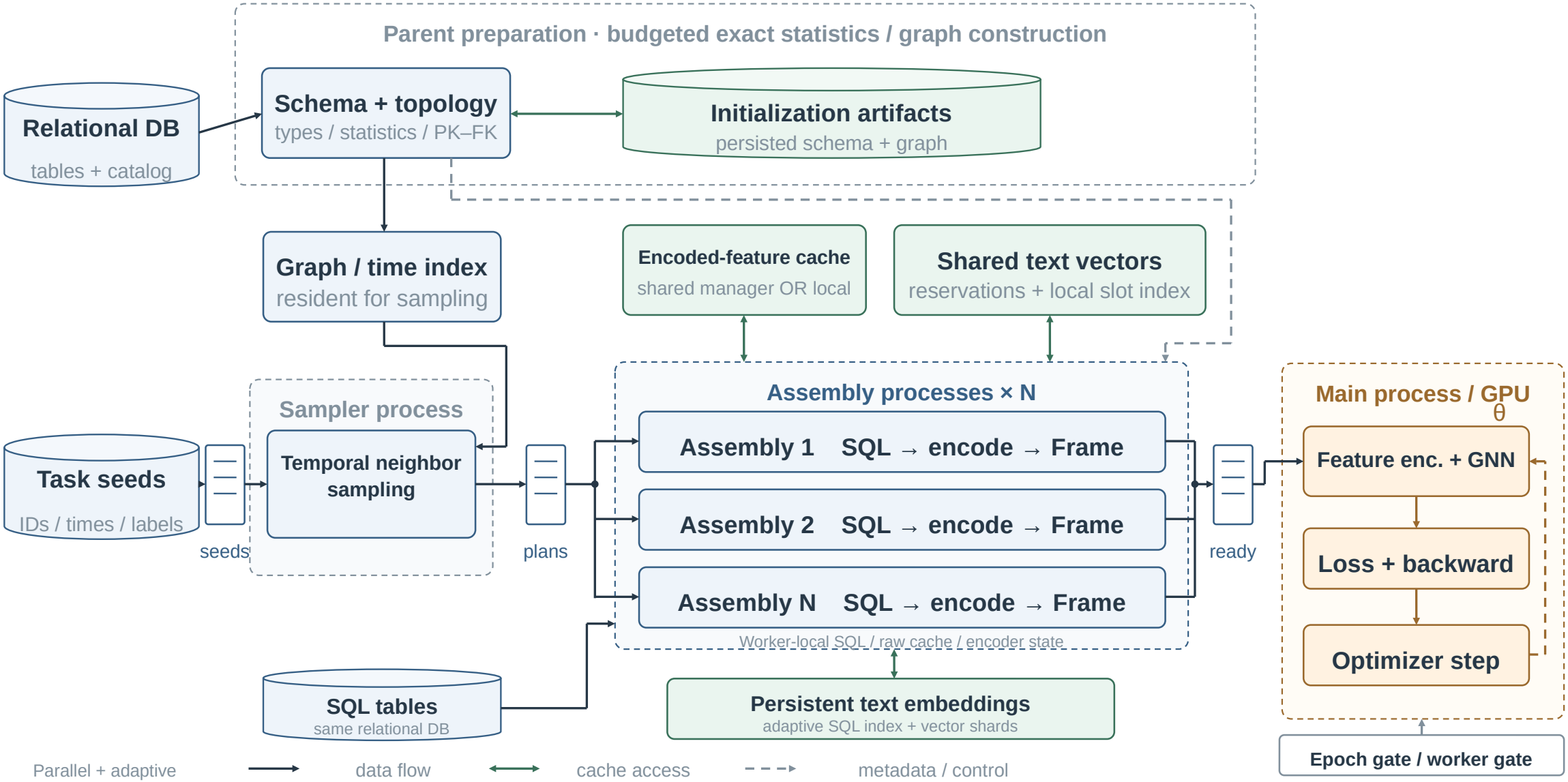
在单进程线程流水线中合并请求，并复用 raw 、 text 与 encoded 特征；训练按序消费。



v1: deduplicated feature windows, layered in-process caches, overlapping fetch / encoding threads, and restored per-batch inputs.

(d) v2 | 多进程与自适应缓存训练架构

assemble 成为供给瓶颈后， v2 将采样、组装、训练分进程执行。



v2 upgrades v1 after assembly becomes a bottleneck: sampling, assembly and training run in separate processes connected by bounded queues.